

CBSE Test Paper 05
Chapter 4 Motion in A Plane

1. A cricketer can throw a ball to a maximum horizontal distance of 100 m. How much high above the ground can the cricketer throw the same ball? **1**
 - a. 76 m
 - b. 50 m
 - c. 65 m
 - d. 60 m
 2. The angle between $A = \hat{i} + \hat{j}$ and $B = \hat{i} - \hat{j}$ **1**
 - a. 180°
 - b. -45°
 - c. 90°
 - d. 45°
 3. The position of a particle is given by $r = 3.0t\hat{i} - 2.0t^2\hat{j} + 4.0\hat{k}$ Find the velocity and acceleration of the particle. **1**
 - a. $5.0\hat{i} - 4.0t\hat{j}, -3.0\hat{j}$
 - b. $2.0\hat{i} - 4.0t\hat{j}, -2.0\hat{j}$
 - c. $3.0\hat{i} - 4.0t\hat{j}, -4.0\hat{j}$
 - d. $4.0\hat{i} - 4.0t\hat{j}, -3.0\hat{j}$
 4. Which of the following physical quantities a vector? **1**
 - a. temperature
 - b. impulse
 - c. charge
 - d. gravitational potential
 5. A ball is thrown upwards with an initial velocity of 10 m s^{-1} . Determine the maximum height reached above the thrower's hand. Determine the time it takes the ball to reach its maximum height. **1**
 - a. 5.25 m, 0.42 s
 - b. 5.23 m, 1.12 s
 - c. 5.43 m, 0.92 s
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d. 5.10 m, 1.02 s

6. A football is kicked into the air vertically upwards. What is its (a) acceleration, and (b) velocity at the highest point? **1**
 7. Under what condition the three vectors cannot give zero resultant? **1**
 8. Two bodies are projected at an angle θ and $(\pi/2 - \theta)$ to the horizontal with the same speed. Find the ratio of their time of flight. **1**
 9. The greatest height to which a man can throw a stone is h. What will be the greatest distance up to which he can throw the stone? **2**
 10. If unit vectors $\hat{a}$ and $\hat{b}$ are inclined at angle θ , then prove that $|\hat{a} - \hat{b}| = 2 \sin \frac{\theta}{2}$. **2**
 11. A vector $\vec{a}$ is turned through a small angle $\Delta\theta$ without a change in its length. What are $|\Delta\vec{a}|$ and Δa ? **2**
 12. Find the magnitude and direction of the resultant of two vectors A and B in terms of their magnitudes and angle θ between them. **3**
 13. Show that for a projectile fired at an angle of elevation θ , there are two times for which the projectile travels the same vertical distance. Also, prove that the sum of the two times is equal to the total time of flight. **3**
 14. A passenger arriving in a new town wishes to go from the station to a hotel located 10 km away on a straight road from the station. A dishonest cabman takes him along a circuitous path 23 km long and reaches the hotel in 28 min. What is **3**
 - a. the average speed of the taxi?
 - b. the magnitude of average velocity? Are the two equal?
 15. A man can swim at the rate of 5 km/h in still water. A river 1 km wide flows at the rate of 3 km/h. A swimmer wishes to cross the river straight. **5**
 - i. Along what direction must he strike?
 - ii. What should be his resultant velocity?
 - iii. How much time he would take to cross?
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